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Freehand 3D ultrasound imaging of the femoral head and neck

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Keywords

3d, bioengineering, components, circuits, devices and systems, computing and processing, femoroacetabular impingement, freehand ultrasound, imaging, semantic segmentation, signal processing and analysis, transfer learning

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Abstract—Objective: To evaluate the performance of a novel freehand 3D ultrasound system for assessing morphological deformities of the femoral head and neck associated with femoroacetabular impingement syndrome (FAIS).

Methods: The system integrates an off-the-shelf optical tracker with a low-cost point-of-care ultrasound probe. Bone surfaces were segmented from B-mode images using a deep-learning feature pyramid network trained on a publicly available dataset and refined via transfer learning with curated, manually-segmented images. Deterministic algorithms were used to generate bone surface point clouds and meshes. System accuracy was assessed by comparing reconstructed bone surfaces to computed tomography (CT) in four cadaveric hips and one in vivo hip. The system's feasibility for clinical imaging was evaluated in five healthy volunteers.

Results: The ultrasound reconstructions achieved agreement with CT, within-subject repeatability and within-subject reproducibility of 0.4-0.6mm. Given that clinically significant bone deformities associated with FAIS are ≥ 2 mm, the system's performance is sufficient for reliable morphometric assessment. In vivo imaging demonstrated feasibility for clinical use.

Conclusion: The combination of 3D optical tracking, ultrasound imaging, and deep-learning based segmentation using models trained on public datasets and enhanced by transfer learning can produce bone surface maps of the femur head with sufficient accuracy to resolve clinically significant femoral head deformities. The system is suitable for clinical deployment.

Significance: Freehand 3D ultrasound appears to be a viable alternative to CT for assessment of femoral head morphometry in patients with FAIS.

Index Terms—freehand ultrasound, 3D, femoroacetabular impingement, semantic segmentation, transfer learning, imaging

I. INTRODUCTION

FREEHAND 3D ultrasound is an imaging modality that combines conventional 2D B-mode ultrasound scanning with a system for tracking the 3D pose (location + orientation) of the ultrasound probe in space[1]. When combined with an effective semantic segmentation algorithm capable of reliably identifying a tissue boundary of interest and methods for performing 3D volumetric reconstruction, the technology offers a low-cost and radiation-free means of generating large field 3D images of tissue. The method has previously been successfully

applied to 3D imaging of spine[2], kidney[3], liver[4] and breast[5] among other applications. While freehand ultrasound techniques have been investigated for nearly 30 years[6], their clinical application has historically been limited by the shortcomings of available automated ultrasound image analysis algorithms[1]. Traditionally, obtaining high quality 3D reconstructions of most tissue boundaries required a significant amount of manual intervention in segmentation, making the approach incompatible with real-time use or widespread clinical adoption. However, recent developments in automated segmentation based on deep-learning neural networks and the availability of large, high quality, publicly available training sets have resulted in massive improvements in the performance and reliability of ultrasound image segmentation[7], [8].

A recent study investigated the use of neural networks for segmentation and 3D freehand ultrasound reconstruction of the femur in presurgical planning for pediatric patients with hip dysplasia[9]. The study used three different U-Net[10] segmentation model variants and a DeepLabv3+[11] model trained on approximately 21,000 proprietary ultrasound images[12] of the hip bones. Point clouds were extracted and a mesh was rendered using Poisson surface reconstruction[13]. The model was applied successfully to a phantom with a reported accuracy of 0.8mm. Bone surface reconstructions were also obtained in 8 patients, but no quantitative assessment was made of 3D reconstruction accuracy in human anatomy.

In the present study we investigate using freehand 3D ultrasound for imaging the femoral head and neck in adults with the ultimate aim of supplementing or supplanting CT imaging in the planning of arthroscopic surgeries to treat femoroacetabular impingement syndrome (FAIS)[14], a common cause of hip pain. CT imaging is an important tool in the assessment of deformities of the femoral cam (the junction between femoral head and neck) which is the most common cause of impingement between the femur head and acetabulum in FAIS. CT is used to plan surgical resections to restore the cam to a normal, non-impinging shape. As compared to CT, freehand ultrasound offers reduced radiation exposure, more flexibility in positioning the joint, lower cost, and the ability to obtain 3D measurements in the clinic. In contrast to previous

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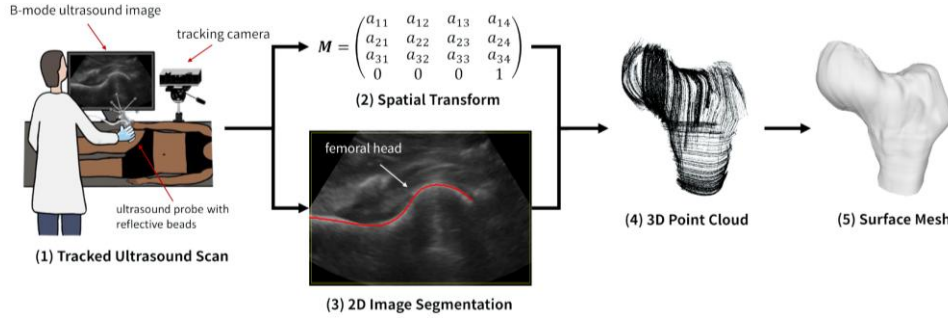


Fig. 1. (a) System diagram showing the Optitrack Duo optical tracker which generates transformation matrices characterizing the pose of an array of tracking beads attached to the ultrasound probe. The ultrasound probe itself generates a stream of 2D B-mode images. Both the transformation matrices and the B-mode images are received by a host PC which segments the images, applies the transformation matrices to situate bone surface contours in virtual 3D space and computes a best-fit surface mesh to the point cloud formed by the contours. (b) Photo showing imaging of a cadaveric hip by the authors using the L12-4 probe.

work on freehand ultrasound of the hip, our study makes use of AI segmentation models trained on a publicly-available rather than a proprietary dataset, namely the UltraBones100k dataset[15] consisting of over 100,000 segmented ultrasound B-mode images of human tibia, fibula and foot. Because the UltraBones100k dataset was acquired on different anatomy than that on which we intend to use it (i.e. tibia and fibula rather than femur head) we employed a transfer learning approach to enhance the model's performance in segmenting the hip bones, using a small number of manually segmented images selected to specifically address observed shortcomings with a model pretrained on the UltraBones100k dataset. We applied this segmentation model within a custom freehand 3D ultrasound system incorporating a low-cost off-the-shelf optical tracker, a low-cost off-the-shelf point-of-care ultrasound probe, and custom image processing software. Bone surface segments in each B-mode image were placed in virtual 3D space to form a point cloud and a custom meshing algorithm based on the radial basis function method was applied to produce 3D reconstructions of the bone surface of the femoral head and neck covering the cam region critical for FAIS assessment.

II. METHODS

Figure 1 provides an overview of proposed freehand 3D ultrasound system work. Figure 1a depicts the acquisition and processing pipeline. Figure 1b shows a photo of the system being used to image a cadaveric hip by two of the co-authors. Supplementary materials 1 contains a video showing the data collection, segmentation and reconstruction processes. Specific components of the system are described below.

A. Tracking system

The system employs an Optitrack Duo (Naturalpoint Inc, Corvallis, OR, USA) optical tracking system to measure the 3D pose of the ultrasound transducer which is fitted with a custom 3D tracker passive marker bead holder designed for continuous tracking over ± 70 degrees of rotation about the long axis of the transducer face and ± 30 degrees rotation about the short face. The tracker samples the pose of the ultrasound at a specified rate of up to 120Hz with specified accuracy ≤ 0.5 mm. After appropriate calibrations, the device reports a transformation matrix from the coordinate axes of the tracker cameras array to the coordinate system of the B-mode image plane.



Figure 2. Optitrack Duo optical tracker and Phillips Lumify Ultrasound probes with custom optical marker bead holders. Left: Lumify L12-4, Right: C5-2.

The system can be used with either one of two off-the-shelf point-of-care ultrasound probes, the Philips Lumify L12-4 linear probe or the Philips Lumify C5-2 curvilinear probe (Philips Medical Systems, Eindhoven, Netherlands), each of which had its own custom, 3D-printed holder for reflective marker beads (Figure 2). The reflective marker bead holders were designed such that the beads were always visible to the camera throughout the intended trajectory of the probe. An asymmetric arrangement of beads was used to avoid ambiguity in the probe's position and rotation. We calibrated each of the probes to co-register the local ultrasound image coordinate system to the tracker's global coordinate system. This was achieved by imaging the tip of a stainless steel stylus with integrated marker beads (Synaptive Medical, Toronto Canada). The stylus tip was placed in a water bath in the image plane of the ultrasound probe, and the stylus tip's location within the ultrasound image was recorded simultaneously with the stylus' pose in the tracker's coordinate system. This measurement was enabled by the fiducial registration module [16] in the software package 3D Slicer v5.9.0[17]. The registration was performed at 6 locations in the ultrasound window and a least-squares fit was used to calculate the relative transformation matrix between the ultrasound image coordinate system and the optical tracker's coordinate system. This process yielded a probe-to-image transformation matrix that was used to orient and place each B-mode image plane in virtual 3D space using the tracker's estimate of the probe's pose.

A separate calibration was performed for each discrete imaging depth supported by each probe—30mm, 35mm, 40mm, and 50mm for the L12-4 probe and 60mm, 70mm, 80mm, and 90mm for the C5-2 probe. Although hip imaging was successfully performed with both probes, the images obtained

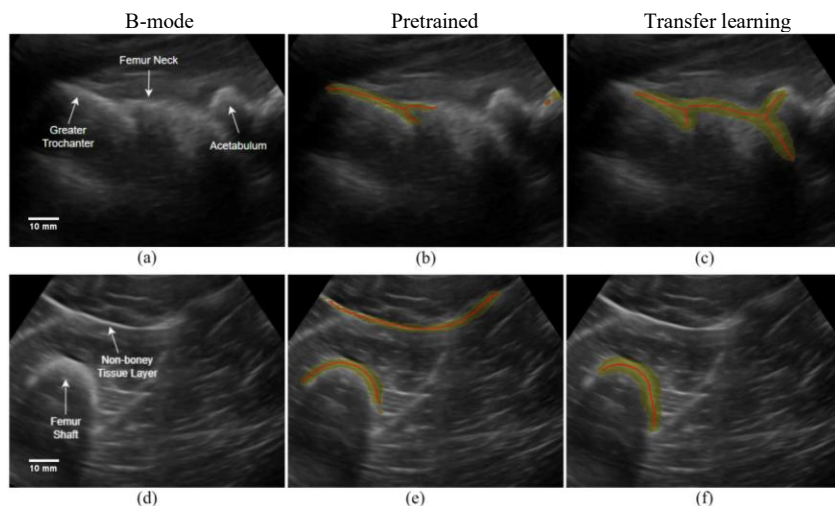


Figure 3. A set of typical B-mode image frames from among those collected in the study showing the segmentation of the bone surface. (a) B-mode frame containing the greater trochanter, femur neck and acetabulum. (b) The segmentation model pretrained on the Ultrabones100k dataset correctly segments the greater trochanter but fails to segment the femur neck or acetabulum. (c) Following enhancement of the model with transfer learning all three structures are correctly segmented. (d) A B-mode frame containing the femur shaft and a strong echo from overlying soft tissue. (e) The pretrained model incorrectly segments the overlying soft tissue as bone as well as the femur. (f) Following transfer learning only the femur is segmented.

with the C5-2 probe were superior owing to the greater penetration depth which allowed the bone surface to be identified from all imaging angles. This was not possible with the L12-4 probe where the bone surface was sometimes lost below the bottom of the image even on the deepest setting of 50mm. All images reported in this study were therefore obtained with the C5-2 probe set to a depth of 80mm.

B. Ultrasound data collection system

The Philips Lumify ultrasound probes are designed to be used with Android tablets or phones, but because this application required collecting and processing a stream of image data from the probes, we instead connected them to a personal computer (PC) running a virtualized Android emulator (Genymotion, Paris, France). A virtual Android tablet screen including the ultrasound B-mode image was captured and encoded using a screen capture server (FFMPEG[18]) and streamed through a real-time media server (MediaMTX[19]) over the real-time streaming protocol (RTSP)[20]. Simultaneously, the transform matrices representing the current pose of the ultrasound probe as reported by the optical tracker were streamed from Optitrack's Motive software interface. These two real-time data streams (B-mode images + transform matrices) were received by an instance of the software Plus Server[21] which bundled and restreamed the data over the OpenIGTLink protocol[16], a light-weight TCP/IP-based protocol designed for real-time data transfer in image-guided therapy. 3D Slicer was used to receive and record these streams using the SlicerIGT client module and the Sequence module and to save them to a file in 3D Slicer's medical reality bundle (.mrb) format. SlicerIGT was also used to apply the transformation matrices to the collected B-mode images to orient and display them in virtual 3D space and to generate a coarse volumetric reconstruction from intensity threshold segmentation to guide probe alignment during images acquisition. At the end of each

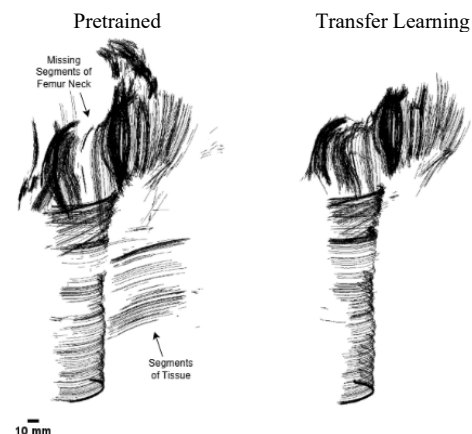


Figure 4. Comparison of pointcloud generated from pretrained model (left) and transfer learning model (right). Each black curve is the segmented bone segment from a single B-mode image. The effects of transfer learning can be seen as a reduction in segmentation of structures other than the femur, longer continuous segments and more coherence bone surface contours when the same region is segmented from multiple directions.

data collection session, image and transform data was exported to a .mrb file and then loaded into custom segmentation and surface reconstruction software written in Python.

C. Segmentation

To segment images, we implemented a custom semantic segmentation model employing a Feature Pyramid Network (FPN) architecture[22] in which the multi-scale feature maps are extracted from the successive stages of a ResNet34 backbone network. The FPN constructs a top-down pathway with lateral connections, where high-level semantic features are upsampled and combined with lower-level spatially detailed features. This produces a set of multi-resolution feature maps that maintain strong semantics at all scales, enabling robust detection of structures of varying size[22]. The model was implemented in the pytorch framework[23] and trained on the UltraBones100k dataset[15], a set of over 100,000 high-quality, segmented B-mode ultrasound images of the lower leg with segmentation obtained from intensity-thresholding of co-registered high resolution CT images of the tibia, fibula and foot in 14 human cadaveric lower legs. All images were resized to 256×256 pixels prior to being used for training. The segmentation model could either be applied in post-processing to datasets saved from 3D Slicer or to the live datastream broadcast from Plus Server in real time (see video in Supplementary Materials). When segmenting images from this stream in real time, segmentation could be inferred and displayed at 10FPS which proved useful in assisting the user to achieve accurate targeting and alignment of the probe during data acquisition.

The output from the segmentation model was a 2D segmentation map of the bone surface which is typically thickened to 1-2mm in B-mode images of bone taken at 5MHz because of the finite axial and lateral resolution of the beam and the strength of the echo as compared to the surrounding soft

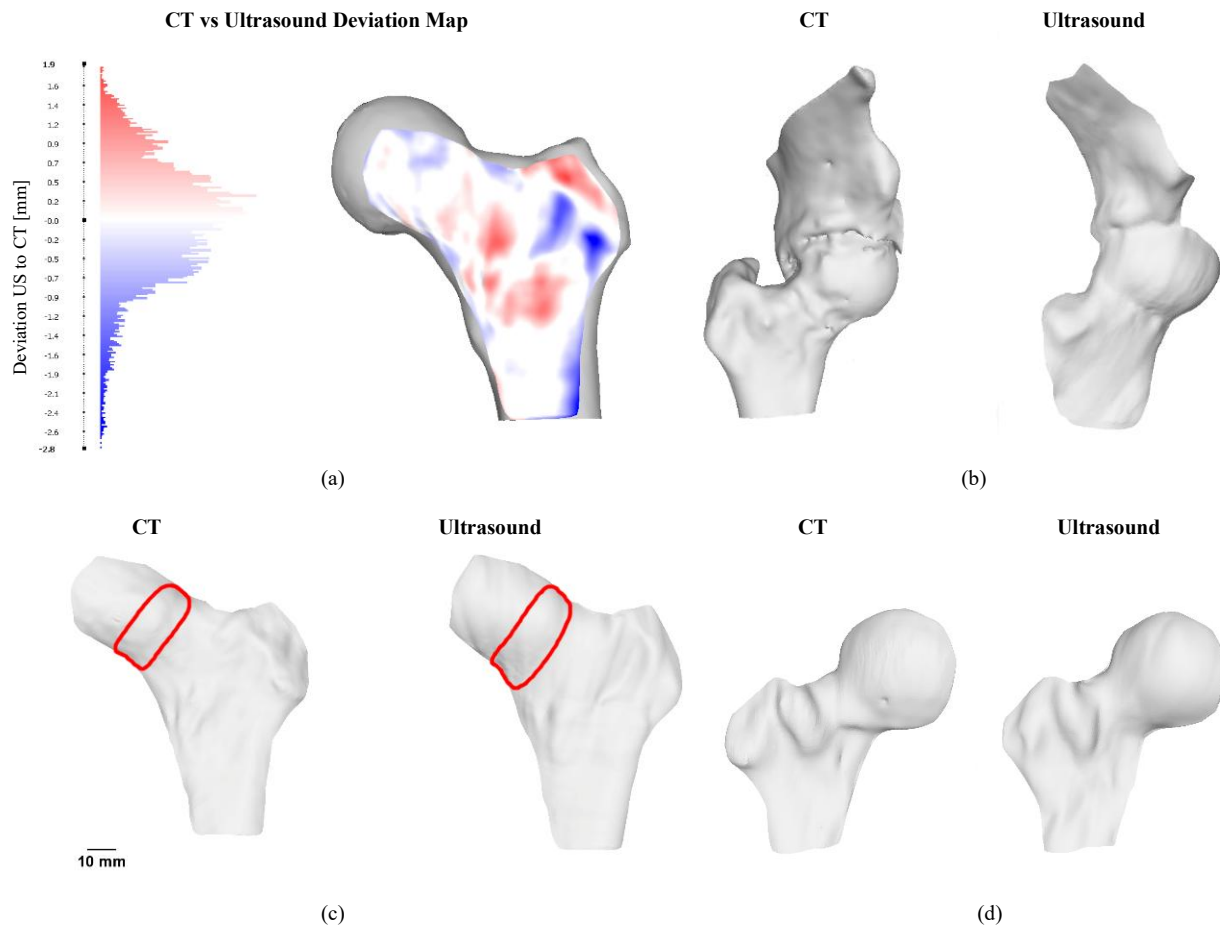


Figure 5. 3D reconstructions of the femur head and neck from freehand 3D ultrasound data with comparative CT images of the same bone. a) Signed deviation between CT and ultrasound for the left hip of cadaveric subject 1 represented as a color map overlay. b) Qualitative comparison between CT and ultrasound for the right hip of cadaveric specimen 2. c) Qualitative comparison between CT and ultrasound for the left femur of cadaveric specimen 1. This bone had a femoral cam deformity which is highlighted in red. It can be readily identified by the lack of a normal concavity in the femoral neck region. d) Qualitative comparison between CT and ultrasound, for the right hip of the healthy volunteer.

tissues[24]. To estimate the surface location as a 1D curve, the 2D segment was skeletonized using the `scikit-image skeletonize()` function[25] followed by pruning with the `deprune` library[26]. The output from this step was a 1D contour delineating the estimated location of the bone surface in each B-mode image. This contour was fit to a continuous B-spline which was used to resample the contour to a constant number ($N=301$) of equidistant points along the contour. Examples of bone contours produced in this way can be seen in Figure 4.

When applied to our femoral head and pelvis ultrasound images, models trained on the UltraBones100k dataset exhibited high selectivity for bone surfaces. However, the segments often exhibited geometric distortion relative to true bone surfaces, misidentification of overlying tissue boundaries as bone, and incomplete segmentation of bone surfaces in frames with multiple bones. Postulating these errors to be due to differences between the morphology of the tibia and fibula in the training set and the femur head and acetabulum in the measurement set, we manually segmented 850 images containing hip bones collected in healthy volunteers with the C5-2 probe. The images were selected to exhibit the types of

errors in the bone surface segmentation that were observed with the pretrained model. This set of manually segmented images was then augmented five-fold by applying random geometric and photometric transformations[27]. The pre-trained FPN model was updated by applying transfer learning[28] on these selected images over 100 epochs, with a batch size of 32 and initial learning rate of 10^{-4} . Validation was performed on a set of 150 additional manually segmented images that were not part of the training set. A terminal validation Dice score of 0.789 was achieved on this validation set by the fully trained model. The transfer learning-enhanced model was then applied to prospectively collected datasets of freehand ultrasound images.

Figure 3 shows two examples of B-mode frames from *in vivo* data collection that were imperfectly segmented by the pretrained FPN model and were subsequently correctly segmented by a model enhanced by transfer learning. Figures 3a, 3b and 3c show a case where the pretrained model failed to segment the acetabulum and femoral neck, in a frame containing acetabulum, femoral neck and greater trochanter surfaces, and the subsequent successful segmentation of these structures following transfer learning. Figure 3d, 3e and 3f show a case where the pretrained model incorrectly identified

TABLE I
GEOMETRIC COMPARISON OF ULTRASOUND AND CT BONE SURFACES

Subject	Side	ASSD[mm]	RMS distance[mm]	HD ₉₅ [mm]	SD _{0.5mm} [mm]	SD _{1mm} [mm]	SD _{2mm} [mm]
Cadaver 1	Right	0.815	0.616	1.774	0.530	0.797	0.980
Cadaver 1	Left	0.395	0.309	0.796	0.774	0.994	1.000
Cadaver 2	Right	0.505	0.406	1.023	0.688	0.951	1.000
Cadaver 2	Left	0.478	0.394	0.863	0.673	0.978	1.000
Healthy volunteer	Right	0.582	0.466	1.216	0.584	0.917	1.000
Mean		0.555	0.438	1.13	0.650	0.927	0.996
SD		0.16	0.11	0.39	0.095	0.079	0.009
Repeatability (Cadaver 1 Left) [mean±SD]		0.46±0.13	0.57±0.15	1.15±0.29	0.63±0.14	0.91±0.08	0.999±0.002
Reproducibility (Cadaver1 Left) [mean±SD]		0.50±0.17	0.62±0.20	1.20±0.41	0.57±0.17	0.89±0.11	0.994±0.014

an overlying soft tissue boundary as a bone surface and the subsequent non-segmentation of this structure following transfer learning.

D. Point cloud generation

The B-spline describing each contour was used to obtain normal vectors oriented towards the proximal direction in the image plane at each resampled contour point. Each point and normal was then transformed from the 2D coordinate frame of the B-mode image into the global 3D coordinate space of the tracker by applying the transform matrix obtained from the tracking system. This produced a point cloud of oriented points lying on the virtual bone surface and pointing anatomically laterally from it. A typical 3 to 5 minute scan would produce a point cloud with between 250,000 and 500,000 points such as those shown in Figure 4.

Figure 4 also shows the improvement in segmentation performance following transfer learning. The transfer-learning-trained model exhibits a lower incidence of incorrectly segmented contours (fewer false positives) and a higher incidence of correctly segmented bone structures (fewer false negatives). For the sample dataset shown in Figure 4, misidentification of overlying soft tissue structures as bone surface was reduced from 4.4% of 3D points to 1.6%. At the same time, the number of correctly segmented bone points within the region of interest relevant to assessing cam deformities (red boxes in Figure 5) increased by 22.5% following transfer learning. Additionally, the coherence of overlapping contours over the same region of bone was noticeably improved as can be seen along the femur shaft in Figure 4.

E. Meshing

We developed a meshing algorithm employing the radial basis function method [29], [30] to interpolate a continuous surface of best fit to the point cloud describing the bone surface. The algorithm employed an open source implementation of the fast multipole method approach to radial basis fitting[31]. Poisson Disk resampling was applied to reduce point cloud density in high-density areas to achieve at most 300,000 total points in the cloud. Mesh generation on a Dell Aurora R16 workstation with an Intel Core i9-14900F (24- core, 68 MB L3 cache, 5.8 GHz) CPU, an NVIDIA GeForce RTX 4070 Ti (16 GB) GPU, and 32 GB DDR5-5600 memory running Windows 11 required 780

seconds (13 minutes) for a 300,000-point cloud. The implementation was CPU-only and single-threaded and so runtime could likely be improved significantly through parallelism.

F. Geometry agreement measures

To assess the accuracy of the femoral head and neck surfaces reconstructed from freehand ultrasound data, we compared each reconstructed surface against ground truth CT image data. To assess repeatability, we performed pairwise comparisons between surfaces generated across multiple freehand ultrasound acquisitions on the same hip. To assess reproducibility, we compared surfaces generated from multiple freehand ultrasound acquisition of the same hip to CT of that hip. In each of these assessments, we calculated each of the six measures of geometric agreement between surfaces described below:

Average symmetric surface distance (ASSD)[32] is the mean of the closest-point distances computed in both directions between two surfaces, providing a single measure of their overall average geometric discrepancy. It is expressed by

$$ASSD = \frac{\sum_{i=1}^{N_A} d_A(i) + \sum_{j=1}^{N_B} d_B(j)}{N_A + N_B}$$

Where $d_A(i)$ is the absolute distance from the i th vertex on mesh surface A to the nearest point on surface B and $d_B(j)$ is the absolute distance from the j th vertex on mesh surface B to the nearest point on surface A . N_A and N_B are the total number of vertices on meshes A and B .

RMS surface distance[32] is the square root of the mean of the squared closest-point distances (in both directions) between two surfaces, providing a variance-weighted measure of their geometric discrepancy:

$$RMSD = \sqrt{\frac{1}{N_A + N_B} \left(\sum_{i=1}^{N_A} d_A^2(i) + \sum_{j=1}^{N_B} d_B^2(j) \right)}$$

The 95th-percentile Hausdorff distance (HD95)[33] is the Hausdorff metric defined as the 95th percentile of the closest-point distances between two surfaces, maximized over the two directions:

$$HD_{95} = \max(\text{percentile}_{95}[\{d_A(i)\}_{i=1}^{N_A}], \text{percentile}_{95}[\{d_B(i)\}_{i=1}^{N_B}])$$

Finally, the surface Dice score[34] at a tolerance τ is the fraction of surface points from both surfaces that lie within a distance τ of the other surface.

$$SD_{\tau} = \frac{|\{d_A(i) \leq \tau\}_{i=1}^{N_A}| + |\{d_B(j) \leq \tau\}_{j=1}^{N_B}|}{N_A + N_B}$$

Where $|\{d_A(i) \leq \tau\}_{i=1}^{N_A}|$ denotes the number of vertices on A lying within a distance τ of surface B and $|\{d_B(j) \leq \tau\}_{j=1}^{N_B}|$ denotes the number of vertices on B lying within a distance τ of surface A .

III. RESULTS

A. Imaging performance

To test the system, we applied it to imaging the femoral head and neck in four human cadaveric hip specimens and a healthy volunteer who had received a CT scan for acute appendicitis, unrelated to this study. All procedures were conducted under oversight from the Nova Scotia Health Research Ethics Board and consent was obtained from all subjects. The four cadaveric hip specimens came from two donors, both male and aged 90 and 86 at the time of death. Prior to performing ultrasound imaging, computed tomography (CT) imaging was performed on the cadaveric hips with a Siemens Somatom Definition AS+ CT scanner operating at a current of 220 mA and voltage of 120 kVp with 0.6mm slice thickness and 0.8mm pitch. Although performed using a clinical scanner, the intensity and resolution of the scan were higher than is typical for clinical scans, taking advantage of the fact that for a cadaveric specimens scan time and radiation dose can both exceed normal clinical limits. The healthy volunteer was imaged with a GE Revolution-series scanner in a contrast-enhanced axial abdominal/pelvic scan at 2mm slice thickness and spacing at a voltage of 120kVp and current of 165mA.

During ultrasound scanning, the specimens were oriented such that the anatomical lateral right or left direction was pointing vertically up as would be the case during a surgery in which the patient is placed in a lateral recumbent position.

Fortuitously for this study, one of tissue donors (cadaver 1) had pronounced cam deformities in both hips which were readily visible in both the ultrasound reconstructions and in the CT scans (see red outline in Figure 5c), while the other donor and the healthy volunteer did not. The cam deformity can be recognized as the absence of a normal concave contour at the femoral head-neck junction characterized by a bony prominence or flattening on the anterosuperior aspect of the femoral neck.

To assess the accuracy of the ultrasound reconstruction we co-registered the ultrasound and CT scans in 3D Slicer using rigid transformations obtained by matching four manually selected anatomical landmarks on the femoral head, femoral neck, femoral shaft and greater trochanter. This was followed by fine co-registration using the iterative closest point (ICP)[35] algorithm from the SlicerRT module. Once the two datasets were co-registered, the CT scan was cropped to the region where it overlapped with the ultrasound scan and the

surface distance agreement metrics listed in the previous section were calculated between the ultrasound 3D reconstruction and the CT. The results can be found in Table I. Measures of average distance (ASSD, RMS distance) were in the range of 0.4mm to 0.6mm. H95, a robust measure of worst-case disagreement ranged from 0.8 to 1.7mm. Surface Dice scores show that roughly 70% of points on the ultrasound-derived surface lie within 0.5mm of the CT surface, 93% of points lie within 1mm of the CT surface and >99% of points lie within 2mm of the CT surface. It should be noted that for all agreement measures, the worst values were for cadaver 1, right hip. Since this was the first hip measured in the series, data collection methods (e.g. probe trajectory, probe pressure, sample positioning) had not yet been optimized. Lessons learned from that first sample informed procedural improvements that were applied in later samples which resulted in higher accuracy.

To verify system reliability and repeatability, multiple freehand 3D ultrasound scans were performed on one of the hips (cadaver 1, left hip). Reproducibility was assessed by comparing each of six scans of the same hip to the registered CT of that hip. Repeatability was assessed by making pairwise comparisons between five ultrasound scans of the hip without reference to the CT. Mean repeatability and reproducibility obtained from these tests was within one standard deviation of the mean accuracy measured across the four cadaveric hips and the healthy volunteer. The close agreement between the accuracy and repeatability/reproducibility suggests that random scan-to-scan error dominates over any systematic error or bias of the freehand 3D ultrasound acquisition and reconstruction process relative to CT. This was, in spite of no correction having been made for tissue aberration or variability in speed of sound in tissue as compared to the water bath where calibration was performed.

We also assessed the ability of ultrasound to visualize the region of the femoral head and neck in which cam resection to treat FAI is typically performed, shown as a red box in Figure 5(c). We found that 3D freehand ultrasound could fully cover the bone surface in the region of the femoral head and neck where cam deformity is located in all 5 of the imaged hips.

B. Live imaging

To assess the potential clinical imaging performance of the system, we also collected 3D femur head reconstructions in five healthy volunteers without corresponding CT scans. High quality images were collected in all five subjects. Two of the subjects had identifiable femoral cams similar to the one observed in the cadaver hip of Figure 5c, while the other three had normal femur head and neck anatomy similar to that of the healthy volunteer of Figure 5d. Although we did not have ground truth CT to compare against the ultrasound reconstructions for these subjects, there was no apparent decrease in the quality of the surface reconstruction when imaging these in vivo hips as compared to the cadaveric hips. Thus, motion artefact does not appear to present a significant impediment to imaging, even when no patient restraints are

used.

Imaging was attempted in both supine and lateral recumbent positions. While high quality bone surface images could be obtained of the anterior aspect of the femur in both positions, the superolateral aspect of the femoral neck could only be accessed in the recumbent position, making this the preferred position for imaging. No significant impediments to clinical deployment of the system were noted across these five imaging sessions.

IV. DISCUSSION

Comparison of the CT and ultrasound scans in the cadaveric sample indicates a mean accuracy of about 0.5mm across four human cadaveric hips and one healthy volunteer. This measured accuracy may be compared with the typical prominence of the cam deformity in FAIS patients of 2-5mm, which can reach 7-8mm in severe cases[36], [37]. The comparison indicates that the current imaging accuracy of the system will likely be adequate to detect, assess and plan the resection of cam deformities in the FAIS patient population when this system is deployed clinically.

In scanning both the cadavers and live subjects we deliberately moved the probe very slowly to obtain a high density of B-mode images, taking 3-5 minutes per scan. This slow scan rate was necessitated by the low B-mode framerate of between 20-30 frames per second limited by the high computational load of running the Lumify app through a virtual Android emulator. Integration of the tracker and system software with a standard ultrasound system capable of achieving frame rates of >30FPS could realistically allow faster scanning and acquisition of 3D volumes with similar density to those collected in this study in as little as one minute.

A difficulty noted in previous freehand 3D ultrasound systems is the susceptibility of image quality to patient motion[9]. In this study, however, we did not observe a significant degradation in quality in going from the cadavers to awake patients, despite not using any restraints. Having patients lie recumbent on a bed appeared adequate to avoid any significant motion artefacts over the duration of an imaging session. The ease of obtaining artefact-free images will ultimately be an important factor in the clinical acceptability of this system, and so these early results are encouraging.

V. CONCLUSION

Freehand 3D ultrasound imaging can provide high-quality 3D reconstructions of the surface of the femur head suitable for detecting and assessing cam deformities associated with femoroacetabular impingement syndrome. As shown in this study, freehand ultrasound incorporating a deep-learning segmentation model pretrained on publicly available datasets enhanced by transfer learning on a small number of anatomy-specific images can offer accuracy within 0.5mm of clinical CT at a much lower cost and without the safety and logistical concerns associated with ionizing radiation. Testing in healthy controls indicates that the system can produce high quality

images in live patients without the use of restraints and that the region of interest for cam deformity assessment is accessible in most subjects. These results suggest that freehand 3D ultrasound may be a viable alternative to CT for clinical 3D imaging of FAIS patients with cam deformities.

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